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# Module 04 - Exploratory Data Analysis
## Group 4 Contribution Journal
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**Course:** ITAI 1371 - Introduction to Machine Learning
**Assignment:** Lab 04 - Exploratory Data Analysis
**Group:** Group 4
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## Patrick Rostand Gandjouon Tchassem
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My contribution to the Module 04 lab was working on the Student Experimentation section. I created a countplot to compare the port where passengers boarded the Titanic with their survival. I also created a boxplot to compare passenger fares with survival.

I reviewed both graphs to understand the patterns in the data. This helped me learn how visualizations can show relationships that may not be easy to see by only looking at numbers. Through my contribution, I practiced using Python, Matplotlib, and Seaborn and gained a better understanding of how Exploratory Data Analysis is used before building a machine learning model.

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## Collaborator 2 - [Saimi Manasiya]
L04 Individual Contribution Journal
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ITAI 1371 - Machine Learning
Module 04: Exploratory Data Analysis
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For Lab 04, my contribution focused on exploring and analyzing the Titanic dataset using Exploratory Data Analysis techniques. I worked with the dataset, reviewed its information, and used visualizations to understand relationships between different variables and survival.

I examined patterns involving passenger class, gender, age, port of embarkation, and fare. The visualizations helped me identify important differences between passengers who survived and those who did not. In particular, I observed that passenger class and gender showed strong relationships with survival.

I also completed the experimentation section by creating visualizations to explore additional relationships in the dataset. This helped me practice selecting appropriate plots and interpreting what the results showed instead of only looking at numerical statistics.

Through my contribution, I learned that EDA is more than creating graphs. The important part is understanding what the data and visualizations are telling us and using those observations to make better decisions for machine learning.

Overall, my work contributed to the group's understanding of the dataset and helped me develop a better understanding of how EDA fits into the machine learning workflow.

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## Collaborator 3 - Kenneth Kouokam
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My contribution to the Module 04 lab focused on reviewing data quality and descriptive statistics in the Titanic dataset. I examined the dataset's structure, column types, and missing values to support the group's analysis. The dataset contained 891 passenger records and 12 columns, with 177 missing ages, 687 missing cabin entries, and two missing embarkation entries. I also distinguished numerical measurements, such as age and fare, from identifiers and category labels. This review helped explain which information was available and where conclusions required more care.

I also compared summary statistics to identify values that needed closer attention. For example, the mean fare was about 32.20, while the median was about 14.45, showing how high fares can

pull the average upward. I explained why unusual values should be investigated before being removed and why missing information should be handled separately for each column. My contribution provided context for interpreting the group's graphs and recommendations for preparing the data before future modeling.

Collaborator 4 - [Anavictoria Cavazos]

Personal contribution: My contribution to the assignment for L04 was to communicate with my group and make sure my part was done, which helped prevent the delay in turning in the assignment. I also, on my own, made sure to work on the lab module in Google Colab. My contribution was primarily independent work and seeing how the code works on my own, so that for future labs I can contribute more or take on a specific task to be able to participate more for the benefit of my group.

Collaborator 5 - [Hashim Sayed Hoosini]

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My contribution for Lab 04 primarily focused on working independently to explore and analyze the Titanic dataset due to a scheduling conflict with the rest of the team. My goal was to better understand the data; I explored the dataset for clues, analyzed relationships between variables, and identified patterns involving gender, fare, class, age, and port of embarkation. I imported the necessary libraries such as pandas, matplotlib, and seaborn, ran each code cell within the notebook, and learned how to handle missing values, identify anomalies, and extract insights from the dataset. Additionally, I learned how data visualization helps us better understand data. To further my learning, I completed the experimentation section of the lab using a correlation heatmap of numerical variables.

Overall, my contribution—despite the scheduling conflict—was to work individually to gain a deeper understanding of the concepts and complete the lab within the assigned timeframe, while maintaining communication with the team and following the team lead's direction.